Scrambling in the derivative-singlet channel of complex double-scaled SYK

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Abstract

We evaluate the derivative singlet's out-of-time-order correlator at infinite temperature after removing its conserved component. The joint-moment reduction permits direct use of the established DSSYK four-point channels. A symmetric Jacobi representation turns the selected crossed channel into a positive sum and admits an explicit cutoff bound. The complete same-species correlator includes two uncrossed channels. The known small-deformation scrambling limit then gives growth rate $4\mathcal{J}_s$ in our string time and a logarithmic scrambling scale.

1 A microscopic scrambling observable

Use the full Hermitian Gaussian complex SYK ensemble, standard CAR, $\hbar = 1$, and $\tau_N = \mathbb{E}\, 2^{-N}\,\mathrm{Tr}$ at $\beta = \mu = 0$:

$$H = H_s = \sum_{|I|=|J|=p} G_{IJ} c_I^\dagger c_J, \qquad \mathbb{E} G_{IJ} G_{KL} = w \delta_{IL} \delta_{JK}, \tag{1}$$

$$w = 4^p \frac{\mathcal{J}_s^2}{\lambda} \left(\frac{p!}{N^p}\right)^2, \qquad \lambda = p^2/N, \quad q = e^{-4\lambda}, \quad \sigma^2 = \mathcal{J}_s^2/\lambda. \tag{2}$$

Overlapping index sets are included. First take $N, p \to \infty$ at fixed $\lambda > 0$. With Q fermion number, define

$$A = \sum_i [H, c_i]^\dagger [H, c_i], \qquad O = A - \mathbb{E} A(Q), \qquad B = H^2 - \mathbb{E} H^2(Q), \tag{3}$$

$$X = \frac{O - cB}{\sqrt{W_d}}, \qquad c = 2\lambda \frac{1-q}{1+q}, \qquad W_d = 4\mathcal{J}_s^4 \left(\lambda^{-1} + \frac{4q}{1+q}\right). \tag{4}$$

In the limiting tracial state τ, X has unit variance. Our preceding note establishes its joint chord moments with H/σ and fixed-time convergence, with crossing weights $q, r = q^2, \chi = q^4$ for HH, HX, XX.

The new observable and its squared commutator are

$$F_{\mathrm{OTO}}(t) = \tau(X(t) X X(t) X), \qquad X(t) = e^{itH} X e^{-itH}, \tag{5}$$

$$\mathcal{C}(t) = \tau([X(t), X]^\dagger [X(t), X]) = 2\{1 + (1+\chi) G_{r^2}(t) - F_{\mathrm{OTO}}(t)\}. \tag{6}$$

Here $G_r(t) = \tau(X(t)X)$ is the known normalized q-Mehler kernel. Traciality makes F_{OTO} real, but need not make it positive. The commutator is nonnegative and vanishes at $t = 0$. Direct mixed moments give

$$\mathcal{C}(t) = 2\sigma^2 (1-r)^2 (5 + 2r + 3r^2) t^2 + O(t^4). \tag{7}$$

The coefficient is $1280 \mathcal{J}_s^2 \lambda + O(\mathcal{J}_s^2 \lambda^2)$ at small λ.

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2 Evaluation of the existing four-point channels

We use BINT Eqs. (3.9), (4.9), (4.10) [1]; BNR Section 8.1 [2] supplies the complex-model correspondence. Introduce the symmetric Jacobi matrix J on $\ell^2(\mathbb{N}_0)$:

$$J_{n,n+1} = J_{n+1,n} = \sigma\sqrt{\frac{1-q^{n+1}}{1-q}}, \qquad U_\pm(t) = e^{\pm itJ}, \quad D_r = \mathrm{diag}(r^n). \tag{8}$$

Put $u = U_-|0\rangle$ and $v = U_+ D_r u$. Each of the two uncrossed channels in this alternating-time arrangement is

$$\mathcal{U}(t) = v^{\mathsf{T}} v = \langle 0|U_- D_r U_+^2 D_r U_-|0\rangle. \tag{9}$$

The transpose is essential: this expression is not $v^\dagger v$. Its imaginary part vanishes by the symmetry $J \mapsto -J$.

For the selected crossed channel, define

$$f_k = (q;q)_k, \qquad b_i = \frac{(r^2;q)_i}{(q;q)_i}\, r^i, \tag{10}$$

$$V_{li}(u) = \sqrt{b_i}\, r^l \sqrt{\frac{f_{l+i}}{f_l}}\, u_{l+i}, \qquad l, i \geq 0. \tag{11}$$

The symmetric form of the BINT crossed sum is

$$\boxed{\mathcal{R}(t) = \|D_r^{1/2} U_+ V(u)\|_{\mathrm{HS}}^2, \qquad F_{\mathrm{OTO}}(t) = 2\mathcal{U}(t) + \chi\mathcal{R}(t).} \tag{12}$$

This is a rearrangement of the known transfer sum, not a newly derived R matrix. In particular $\mathcal{R}(t) \geq 0$. The factor $\chi = q^4$ is the mutual-probe crossing factor suppressed in BINT Section 4.2; it must be restored when combining all three same-species channels. At zero time $\mathcal{U} = \mathcal{R} = 1$, giving $F_{\mathrm{OTO}}(0) = 2 + q^4$.

Similarity factors. For the nonsymmetric BINT transfer matrix T, set $s_n = \sqrt{[n]_q!}$, so $(T^k)_{ab} = (J^k/\sigma^k)_{ab} s_b/s_a$. The coefficient $P_i^{(l+i)}$ in their Eq. (3.9), together with four such similarity factors, becomes

$$\frac{(r^2;q)_i}{(q;q)_i}\sqrt{\frac{f_{l+i}f_{m+i}}{f_l f_m}}. \tag{13}$$

The remaining powers are $r^{i+n+l+m}$. The n sum is the matrix element of $U_- D_r U_+$, yielding (12). Thus all factors can be checked before taking real times or any small-λ approximation.

For $r = q^2$ the finite product ratio simplifies to

$$\frac{(r^2;q)_i}{(q;q)_i} = \frac{(1-q^{i+1})(1-q^{i+2})(1-q^{i+3})}{(1-q)(1-q^2)(1-q^3)}. \tag{14}$$

The implementation uses logarithms of f_k and this ratio to avoid ill-conditioned factorial similarities.

3 Finite-deformation dynamics

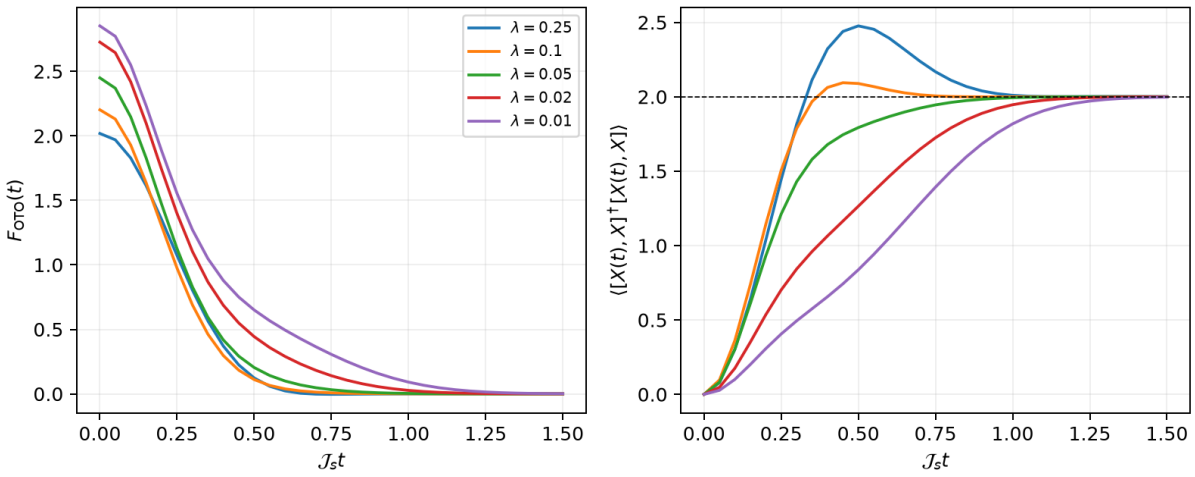


Figure 1. Full OTOC and squared commutator at $\beta = \mu = 0$, with all three channels retained. The dashed line at 2 is a reference; no universal finite-q late-time plateau is inferred from this plot.

λ	$F_{\mathrm{OTO}}(0.5/\mathcal{J}_s)$	$\mathcal{C}(0.5/\mathcal{J}_s)$	L
0.25	0.123101	2.47621	53
0.10	0.110600	2.08900	103
0.05	0.204036	1.79281	190
0.02	0.443462	1.26429	459
0.01	0.650048	0.837141	921

The commutator grows later as λ is reduced. At $\lambda = 0.25$ the sampled OTOC also becomes slightly negative (minimum about -0.00365); the selected crossed channel stays nonnegative, while the uncrossed channels can change sign. This is consistent with (6) and illustrates why the crossed channel alone is not the full same-species correlator.

Independent channel normalization. For every distribution of $0, 2, 4, 6$ background H insertions among the four arcs, direct chord enumeration agrees exactly with the nonsymmetric BINT moment formula at $q = 2/5, 3/4$: 260 moment cases and 780 rational channel comparisons. The coefficient (7) also follows independently from

$$\tau([[H,X],X]^2) = 2\tau(H^2X^4) + 6\tau(HX^2HX^2) - 8\tau(HXHX^3). \tag{15}$$

Four finite-$L = 18$ evaluations at 50 decimal digits use nonsymmetric matrix exponentials and the original index sum. They agree with the positive symmetric implementation to 1.6×10^{-15}. Independent 96/192-node Mehler integrals verify the uncrossed channel at eight time/parameter points to 5.0×10^{-14}. For the main curves, $L + 16$ changes the full OTOC by at most 1.1×10^{-14} at the checked points. These residuals assess floating-point behavior; the infinite-cutoff bound is given separately in Appendix A.

4 The known scrambling limit in string units

Mukhametzhanov [3] gives the long-time small-deformation crossed correlator. Its parameters must be converted before use:

$$\lambda_{\text{source}} = 4\lambda, \qquad \mathcal{J}_{\text{source}} = 2\mathcal{J}_s, \qquad u_{\text{source}} = 2\mathcal{J}_s t, \qquad \Delta = 2. \tag{16}$$

The source sets its coupling to one. Its saddle equation (3.11) implies $\lambda_{L,\text{source}} \to 2$ as $\beta \to 0$. Thus Eqs. (5.20)–(5.23), for the alternating symmetric arrangement, give the scaling variable and limiting crossed function

$$\boxed{z = \frac{\lambda}{2} e^{4\mathcal{J}_s t}, \qquad \mathcal{R}(t) \longrightarrow \mathcal{F}(z) = z^{-4} U(4, 1, 1/z), \quad \lambda \to 0, \ z > 0 \text{ fixed.}} \tag{17}$$

We use the standard normalized Tricomi function, equivalently

$$\mathcal{F}(z) = \frac{1}{\Gamma(4)} \int_0^\infty \mathrm{d}y \, \frac{y^3 e^{-y}}{(1+zy)^4}. \tag{18}$$

This normalization gives $\mathcal{F}(0) = 1$ and agrees with the source's Eq. (5.24). The limit is the existing probe result applied to the singlet reduction; it is not a new derivation of the chaos exponent.

z	$\mathcal{F}(z)$	$\mathcal{R}_{.02}$	$\mathcal{R}_{.01}$	$\mathcal{R}_{.005}$
0.025	0.696227	0.887540	0.800063	0.749483
0.050	0.512910	0.673161	0.592868	0.552409
0.100	0.309611	0.411526	0.358189	0.333041

Each column is evaluated at its own $t = \log(2z/\lambda)/(4\mathcal{J}_s)$. These finite-deformation discrepancies shrink over the three values; they are much larger than numerical truncation errors. They are not an estimate of a proven asymptotic remainder. The smallest parameter uses dimensions up to 1559, with $L + 16$ checks.

At leading order in this scrambling approximation, $\chi \to 1$ and the two uncrossed channels disappear; the full OTOC tends to $\mathcal{F}(z)$ and $\mathcal{C}(t) \to 2[1-\mathcal{F}(z)]$. For $z \ll 1$, $\mathcal{F}(z) = 1 - 16z + O(z^2)$. In the overlap window of late string time but small z, this gives

$$\mathcal{C}(t) \simeq 16\lambda e^{4\mathcal{J}_s t}, \qquad t_* = \frac{\log(1/\lambda)}{4\mathcal{J}_s} + O(\mathcal{J}_s^{-1}). \tag{19}$$

This exponential regime does not replace the analytic t^2 onset (7). The rate refers to string time in the specified microscopic state, not yet to a bulk observer's proper time or Gibbons–Hawking temperature.

A A bound on the infinite transfer and chord cutoffs

Let J_L be the L-dimensional Jacobi compression and embed its evolution back into ℓ^2. Write

$$R_H = \frac{2\sigma}{\sqrt{1-q}}, \quad a = |t|R_H, \quad B_d(a) = \frac{a^d}{d!}\frac{1}{1-a/(d+1)}, \qquad d > a. \tag{20}$$

Taylor locality gives vacuum error $\epsilon_0 = 2B_L(a)$ and, for sources supported on $0 \le l < M < L$, propagator error $\epsilon_M = 2B_{L-M}(a)$. The bound is conservative by one or more Taylor orders and applies uniformly for the chosen time interval.

From (14), define

$$C_q = \frac{1}{(1-q)(1-q^2)(1-q^3)}, \qquad K = \frac{C_q}{1-r}. \tag{21}$$

Because $f_{l+i}/f_l \le 1$ and $b_i \le C_q r^i$,

$$\|V(u)\|_{\mathrm{HS}}^2 \le C_q \sum_{l,i\ge 0} r^{i+2l}|u_{l+i}|^2 \le K\|u\|^2, \tag{22}$$

$$\|\mathbf{1}_{l\ge M}V(u)\|_{\mathrm{HS}} \le \sqrt{K}\, r^M \|u\|. \tag{23}$$

The linear map $u \mapsto V(u)$ has norm at most $\sqrt{K}$. Extending the finite vacuum vector by zero makes its V vanish whenever $l + i \ge L$, so the implemented chord-index cutoff is included in the vacuum error. For $\delta = \epsilon_0 + \epsilon_M + 2r^M$,

$$\|U_+V(u) - U_{+,L}V(u_L)\|_{\mathrm{HS}} \le \sqrt{K}\,\delta. \tag{24}$$

The weighted norm in (12) is contractive. Both unweighted norms are at most $\sqrt{K}$, hence

$$|\mathcal{R} - \mathcal{R}_L| \le 2K\delta. \tag{25}$$

Similarly $D_r u$ has tail at most r^M and norm at most one, so $|\mathcal{U} - \mathcal{U}_L| \le 2\delta$. Consequently

$$\boxed{|F_{\mathrm{OTO}} - F_{\mathrm{OTO},L}| \le (4 + 2\chi K)\delta.} \tag{26}$$

The error of G_{r^2} is at most $2\epsilon_0$; combining with (6) bounds the commutator error by $2[2(1+\chi)\epsilon_0 + (4 + 2\chi K)\delta]$.

The code chooses M, L so (26) is below 10^{-10} up to $\mathcal{J}_s t = 1.5$. Actual main-curve bounds are 3.6–4.9 $\times 10^{-11}$. This controls the infinite Jacobi and chord-index tails, including propagation of non-vacuum sources. It does not certify eigensolver roundoff. An independent 70-digit replay verifies all eight chosen cutoffs. Across 155 main points the uncrossed phase-cancellation condition number reaches 411, while cancellation between the three complete channels stays below 1.84. Relative refinement errors and imaginary residuals are saved; near a zero of the OTOC, absolute errors are the appropriate diagnostic.

B Provenance and the next bulk question

The exact kernels in this note come from [1, 2], and the scrambling asymptotics from [3]. The inputs particular to our calculation are the derivative-singlet reduction established in the preceding note, the full same-species channel assembly, its positive crossed-sum implementation, and the explicit error accounting. This is a result for the ensemble-averaged tracial observables; a samplewise self-averaging theorem for this entire time window is not assumed.

Reproducible evidence. The main numerical file stores 155 finite-q time points, 25 dimension refinements, five short-time checks and nine scrambling-scaling points with their own refinements. Parameters, environment and input hashes are recorded. Python 3.14.5, NumPy 2.4.6, Matplotlib 3.11.0 and mpmath are used. The exact channel checker also records its SymPy version. The high-precision representation checker compares the original nonsymmetric sum with the symmetric formula at identical finite L; it does not masquerade as an independent infinite-cutoff certification. The Tricomi values are checked against the normalized gamma integral.

The source PDF of [3] is cached separately and its Eq. (5.20) page was visually inspected to fix the prefactor and time convention. The source sets its coupling to one; retaining that convention is why the factor $1/2$ appears in our z. Source caching does not grant redistribution rights. All work here remains local, without a Git commit, push or archive. Earlier scientific files remain unchanged.

The physical next step. The static-patch program proposes that singlets furnish propagating bulk excitations [4]. We now have both the propagator and a complete microscopic alternating-time four-point observable for one explicit singlet. The next comparison should therefore use a bulk wavepacket four-point observable with stated Rindler time and spatial smearing, not a mass assigned from the chord label $\Delta = 2$. The known microscopic scrambling function supplies a definite target.

Two concrete tasks remain: derive the first finite-λ correction at fixed time (including the previously obtained adjacent-pair connected function), and determine whether the proposed bulk singlet/meson dictionary supplies the needed wavepacket and time identification. Finite de Sitter curvature and an on-shell scattering interpretation require that additional dictionary; the present calculation does not establish a complete holographic duality.

References

[1] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584v2, Eqs. (2.13), (3.9), (4.9), (4.10); Section 4.2 and footnote 8.

[2] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Section 8.1.

[3] B. Mukhametzhanov, *Large p SYK from chord diagrams*, arXiv:2303.03474v3, Eqs. (3.11), (5.20)–(5.24).

[4] S. Miyashita, Y. Sekino and L. Susskind, *DSSYK at Infinite Temperature: The Flat-Space Limit and the 't Hooft Model*, arXiv:2506.18054v3, Sections 5–6.